Exam 1

Graded

Student

Christian Patrick Byrne

Total Points

96 / 100 pts

Question 1

(no title) — **3** / 5 pts

✔ **+ 3 pts** Truth table: 2 rows correct | Rules of Inference: 2 steps wrong

Question 2

(no title) — **10** / 10 pts

✔ **– 0 pts** Correct

Question 3

(no title) — **5** / 5 pts

✔ **– 0 pts** Correct

Question 4

(no title) — **10** / 10 pts

✔ **– 0 pts** Correct

Question 5

(no title) — **4** / 4 pts

5.1 **(no title)** — **1** / 1 pt

✔ **– 0 pts** Correct.

5.2 **(no title)** — **1** / 1 pt

✔ **– 0 pts** Correct

5.3 **(no title)** — **1** / 1 pt

✔ **– 0 pts** Correct

5.4 **(no title)** — **1** / 1 pt

✔ **– 0 pts** Correct

Question 6

(no title) **6** / 6 pts

6.1 (no title) **2** / 2 pts

✔ **− 0 pts** Correct

6.2 (no title) **2** / 2 pts

✔ **− 0 pts** Correct

6.3 (no title) **2** / 2 pts

✔ **− 0 pts** Correct

Question 7

(no title) **5** / 5 pts

✔ **− 0 pts** Correct

Question 8

(no title) **10** / 10 pts

✔ **− 0 pts** Correct

Question 9

(no title) **3** / 3 pts

9.1 (no title) **1** / 1 pt

✔ **− 0 pts** Correct

9.2 (no title) **1** / 1 pt

✔ **− 0 pts** Correct

9.3 (no title) **1** / 1 pt

✔ **− 0 pts** Correct

Question 10

(no title) **10** / 10 pts

✔ **+ 10 pts** Correct

Question 11

(no title) **4** / 4 pts

✔ **− 0 pts** Correct

Question 12

(no title) **10** / 10 pts

✔ **− 0 pts** Correct

Question 13

(no title) **5** / 5 pts

✔ **− 0 pts** Correct

Question 14

(no title) **5** / 5 pts

✔ **− 0 pts** Correct

Question 15

(no title) **2** / 4 pts

✔ **+ 2 pts** Wrong direction of implication or double implication

Question 16

(no title) **4** / 4 pts

16.1 **(no title)** **2** / 2 pts

✔ **− 0 pts** Correct

16.2 **(no title)** **2** / 2 pts

✔ **− 0 pts** Correct

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CsC 144, Spring 2024 Midterm 1

This is a closed book, closed note exam. Please have nothing on your desk except this test and something to write with. DON'T look at your fellow student's exam.

These exams will be scanned into GradeScope, so while you may use the back of your exam as scratch paper, remember that anything written on the back will not be scanned and thus not seen by the graders.

Please wait to open the exam until told to do so. That way everyone will have the same amount of time to work on the exam.

The last page is for you to use as scratch paper. You can detach it. You don't have to turn it in.

Good luck.

Page	Points
1	0
2	15
3	19
4	21
5	18
6	15
7	13

1. (5 pts) Determine whether (¬p ∧ (p → q)) → ¬q is a tautology. Show your work.

$(!p \wedge (p \rightarrow q)) \rightarrow !q$

$\equiv !(!p \wedge (p \rightarrow q)) \vee !q$

$\equiv !(!p \wedge (!p \vee q)) \vee !q$

~~$\equiv !((!p \wedge !p) \vee (!p \wedge q)) \vee !q$~~

~~$\equiv !(!p \vee (!p \wedge q)) \vee !q$~~

~~$\equiv !(!p) \vee !q$~~

$(p \vee (!p \vee q)) \vee !q$

$\equiv (p \vee !p \vee q) \vee !q$

$\equiv T \vee q \vee !q$

$\equiv T \vee T$

$\equiv T$

2. (10 pts) Use proof by cases to show that for any two positive real numbers x and y, |x – y| = max(x, y) – min(x, y).

-1, +2

Case 1: $x \geq y$

if $x \geq y$, then $|x-y| \equiv x-y$ because positive number minus smaller positive number can't be < 0

and $\max(x,y) = x$ or $x=y$

and $\min(x,y) = y$ or $x=y$

thus, $x-y \equiv x-y$

Case 2: $x < y$

1,2

1

if $x < y$, then $|x-y| \equiv |-y+x| \equiv y-x$ because $-y+x$ will always be the difference between a smaller positive minus a larger positive

and $\max(x,y) \equiv y$

and $\min(x,y) = x$

thus, $y-x \equiv y-x$

Thus, for any two positive real numbers,

$|x-y| = \max(x,y) - m(x,y)$

3. (5 pts) Let P(x) be "x is a platypus", F(x) be "x wears a fedora" the domain being all secret agents. Express the statement "There is a secret agent who is a platypus and who wears a fedora." Using the predicates, quantifiers, and logical connectives.

$\exists x \, (P(x) \wedge F(x))$

4. (10 pts) Show that the product of a nonzero rational number and an irrational number is irrational using a proof by contradiction.

If x is a non zero rational and y is irrational, then xy is irrational

To prove by contradiction, assume that the conclusion (xy is irrational) is false and the hypothesis is true.

If the conclusion is false, then xy is rational and can therefore be stated as the ratio of two integers P/q where $q \neq 0$:

$$xy = P/q$$

$$x = \frac{P}{qy}$$

$$xq = \frac{P}{y}$$

But if q is an integer and x is rational then xq cannot be irrational. However, P/y must be irrational because it is an integer over an irrational number (and we assumed the hypothesis is true). Therefore it is a contradiction and if the hypothesis is true, the conclusion must be true

rational closed over multiplication

5. (4 pts) Indicate whether the following sentences are propositions.

a. x+ 5 = 6.
◯ Proposition ◉ Not a Proposition

b. What are you doing?
◯ Proposition ◉ Not a Proposition

c. Perry is a platypus.

◉ Proposition ◯ Not a Proposition

d. Curse you, Perry the Platypus!
◯ Proposition ◉ Not a Proposition

6. (6 pts) Suppose that last semester CSC 144 had an enrollment of 116 students and an average daily attendance of 44 students, CSC 244 had an enrollment of 199 students and an average daily attendance of 66 students, and CSC 120 had an enrollment of 130 students and an average daily attendance of 69 students. Determine the truth value of each of these propositions:

a. CSC 120 had the largest enrollment.

◯ True ◉ False

b. If CSC 120 had the smallest average daily attendance, then CSC 144 had the largest enrollment.

◉ True ◯ False if $F \to F$

c. CSC 144 had the smallest daily average attendance and CSC 244 had the largest average daily attendance.

◯ True ◉ False $T \wedge F$

7. (5 pts) Is the proposition (p ∨ ¬ q) ∧ (¬ p ∨ q) ∧ (¬ p ∨ ¬ q) satisfiable? Justify your answer.

Yes. It's a conjunction of three 2-variable disjunctions. Each 2-variable disjunction has at least one negated variable. Therefore, if all variables = F, all disjunctions will be True and the conjunction will be true. So there is at least one solution and its therefore satisfiabe

I.e., satisfiable by {p:F, q:F}

8. (10 pts) On a strange island there live two kinds of people, Knights who can only tell the truth, and Knaves who can only lie. Every inhabitant is exactly one of these two types. You meet two people: Phineas and Ferb.
   1. Phineas says "Of Ferb and I, only one of us is a knight".
   2. Ferb says "Phineas is a knight."
   State which type each person is if it can be determined. Prove your answer. Don't use a truth table for your proof.

Case 1: Ferb - Knight →

(1) ∴ Phineas - Knight

(2) $\wedge$!(Ferb-Knight $\wedge$ Phinees-Knight) - F

the conjunction of 1 and 2 evaluate to false, so the case is invalid

Case 2: Ferb - Knave →

(1) → !(Phineas - Knight)

(2) $\wedge$!((Phineas-Knight $\wedge$ Ferb-Knave) $\vee$ (Phineas - Knave $\wedge$ Ferb-Knight))

$\equiv$!(Phineas-Knight $\wedge$ Ferb-Knave) $\wedge$!(Phineas-Knave $\wedge$ Ferb-Knight)

$\equiv$!(F $\wedge$ T) $\wedge$!(T $\wedge$ F)

$\equiv$!(F) $\wedge$!(F)

$\equiv$ T $\wedge$ T $\equiv$ T → (2) Evaluates to True when Ferb-Knave

$\boxed{\text{Ferb is a Knave, Phineas is a Knave}}$

Thus, !(Phineas-Knight) $\wedge$ T

Thus, Phineas Knight

9. (3 pts) Let P(x) be x ≥ 5. What are these truth values?

   a. P(8)

   ● True ○ False

   b. P(7)

   ● True ○ False

   c. P(0)

   ○ True ● False

10. (10 pts) Prove that there is no positive integer n such that $10 + n^2 = 100$.

Direct Proof:

$10 + n^2 = 100$

$n^2 = 90$

$n = \sqrt{90}$

Since $\sqrt{90}$ is not an integer, this is not satisfiable for any integer

11. (4 pts) Express the negation of the following statement so that all negation symbols immediately precede predicates.
∀y∃x∃z(T(x, y z) ∨ Q(x, y))

!

Ey Ax Az(!T(x,y,z) ∧ !Q(x,y))

12. (10 pts) Prove that n is even iff 3n + 8 is even.

if n is even, then $3n+8$ is even and if $3n+8$ is even, then n is even

Part 1: assume n is even, so $n = 2k$ for some int k

thus, $3n+8 = 3(2k)+8$

$= 6k+8 = 2(3k+4)$ which is even

Part 2: assume $3n+8$ is even,

Prove by contraposition by assuming that

if n is not even then $3n+8$ is not even.

If n is not even, then n is odd and $n = 2k+1$ for some integer k

Thus, $3n+8 = 3(2k+1)+8$

$= 6k+3+8$

$= 6k+10+1$

$= 2(3k+5)+1$ which is odd

So if n is odd, then $3n+8$ is odd

and therefore if $3n+8$ is even, n is even by contraposition

Thus, (if n is even, then $3n+8$ is even) and (if $3n+8$ is even, then n is even)

≡ n is even iff $3n+8$ is even

13. (5 pts) Use laws of equivalences to show p→(p ∨ q) is a tautology. (You don't need to name the rules you use.)

$p \to (p \vee q)$

$\equiv\ !p \vee p \vee q$

$\equiv\ T \vee q$

$\equiv\ T$

14. (5 pts) Find a counterexample to the following where the domain is all integers.
$\forall x \forall y((x^2 = y^2) \to (x = y))$

$x = -1,\ y = 1$

15. (4 pts) Translate the following statement into propositional logic. You can see the movie only if you are over 18 years old or you have the permission of a parent. Express your answer in terms of m: "You can see the movie," e: "You are over 18 years old," and p: "You have the permission of a parent."

$e \vee p \to m$

If su, f

16. (4 pts) Determine whether the following arguments are valid.

a. Every secret agent wears a fedora. Doof does not wear a fedora. Therefore, Doof is not a secret agent.

Valid　　　Not Valid

b. If you rode the rollercoaster, you had a great summer. Candace didn't ride the rollercoaster. Therefore, Candace did not have a great summer.

Valid　　　Not Valid